formulation of adversarial training (Szegedy et al., 2013; Goodfellow et al., 2014b), which we use in our experiments, trains on both the "clean" examples x and adversarial examples x^{adv}.

We consider three algorithms to generate adversarial examples with bounded ℓ_∞ norm. The first two are *single-step* (i.e., they require a single gradient computation); the third is *iterative*—it computes multiple gradient updates. We enforce $x^{\text{adv}} \in [0,1]^d$ by clipping all components of x^{adv}.

Fast Gradient Sign Method (FGSM). This method (Goodfellow et al., 2014b) linearizes the inner maximization problem in (1):

$$x^{\text{adv}}_{\text{FGSM}} := x + \varepsilon \cdot \texttt{sign}\left(\nabla_x L(h(x), y_{\text{true}})\right) \ . \tag{2}$$

Single-Step Least-Likely Class Method (Step-LL). This variant of FGSM introduced by Kurakin et al. (2017a;b) targets the least-likely class, $y_{\text{LL}} = \arg\min\{h(x)\}$:

$$x^{\text{adv}}_{\text{LL}} := x - \varepsilon \cdot \texttt{sign}\left(\nabla_x L(h(x), y_{\text{LL}})\right) \ . \tag{3}$$

Although this attack only indirectly tackles the inner maximization in (1), Kurakin et al. (2017b) find it to be the most effective for adversarial training on ImageNet.

Iterative Attack (I-FGSM or Iter-LL). This method iteratively applies the FGSM or Step-LL k times with step-size $\alpha \geq \epsilon/k$ and projects each step onto the ℓ_∞ ball of norm ϵ around x. It uses projected gradient descent to solve the maximization in (1). For fixed ϵ, iterative attacks induce higher error rates than single-step attacks, but *transfer* at lower rates (Kurakin et al., 2017a;b).

3.3 A Degenerate Global Minimum for Single-Step Adversarial Training

When performing adversarial training with a single-step attack (e.g., the FGSM or Step-LL methods above), we approximate Equation (1) by replacing the solution to the inner maximization problem in with the output of the single-step attack (e.g., $x^{\text{adv}}_{\text{FGSM}}$ in (2)). That is, we solve

$$h^* = \arg\min_{h \in \mathcal{H}} \quad \mathop{\mathbb{E}}_{(x, y_{\text{true}}) \sim \mathcal{D}} \left[L(h(x^{\text{adv}}_{\text{FGSM}}), y_{\text{true}}) \right] . \tag{4}$$

For model families $\mathcal{H}$ with high expressive power, this alternative optimization problem admits at least two substantially different global minima h^*:

- For an input x from $\mathcal{D}$, there is no x^{adv} close to x (in ℓ_∞ norm) that induces a high loss. That is,

$$L(h^*(x^{\text{adv}}_{\text{FGSM}}), y_{\text{true}}) \approx \max_{\|x^{\text{adv}} - x\|_\infty \leq \epsilon} L(h^*(x^{\text{adv}}), y_{\text{true}})] \approx 0 \ . \tag{5}$$

  In other words, h^* is robust to all ℓ_∞ bounded perturbations.
- The minimizer h^* is a model for which the approximation method underlying the attack (i.e., linearization in our case) poorly fits the model's loss function. That is,

$$L(h^*(x^{\text{adv}}_{\text{FGSM}}), y_{\text{true}}) \ll \max_{\|x^{\text{adv}} - x\|_\infty \leq \epsilon} L(h^*(x^{\text{adv}}), y_{\text{true}})] \ . \tag{6}$$

  Thus the attack when applied to h^* produces samples x^{adv} that are far from optimal.

Note that this second "degenerate" minimum can be more subtle than a simple case of overfitting to samples produced from single-step attacks. Indeed, we show in Section 4.1 that single-step attacks applied to adversarially trained models create "adversarial" examples *that are easy to classify even for undefended models*. Thus, adversarial training does not simply learn to resist the particular attack used during training, but actually to make that attack perform worse overall. This phenomenon relates to the notion of *Reward Hacking* (Amodei et al., 2016) wherein an agent maximizes its formal objective function via unintended behavior that fails to captures the designer's true intent.

3.4 Ensemble Adversarial Training

The degenerate minimum described in Section 3.3 is attainable because the learned model's parameters influence the quality of both the minimization and maximization in (1). One solution is to use

a stronger adversarial example generation process, at a high performance cost (Madry et al., 2017). Alternatively, Baluja & Fischer (2017) suggest training an adversarial *generator* model as in the GAN framework (Goodfellow et al., 2014a). The power of this generator is likely to require careful tuning, to avoid similar degenerate minima (where the generator or classifier overpowers the other).

We propose a conceptually simpler approach to *decouple* the generation of adversarial examples from the model being trained, while simultaneously drawing an explicit connection with robustness to black-box adversaries. Our method, which we call *Ensemble Adversarial Training*, augments a model's training data with adversarial examples crafted on other *static pre-trained* models. Intuitively, as adversarial examples transfer between models, perturbations crafted on an external model are good approximations for the maximization problem in (1). Moreover, the learned model can not influence the "strength" of these adversarial examples. As a result, minimizing the training loss implies increased robustness to black-box attacks from some set of models.

Domain Adaptation with multiple sources. We can draw a connection between Ensemble Adversarial Training and *multiple-source Domain Adaptation* (Mansour et al., 2009; Zhang et al., 2012). In Domain Adaptation, a model trained on data sampled from one or more *source distributions* $\mathcal{S}_1, \ldots, \mathcal{S}_k$ is evaluated on samples x from a different *target distribution* $\mathcal{T}$.

Let $\mathcal{A}_i$ be an adversarial distribution obtained by sampling (x, y_{true}) from $\mathcal{D}$, computing an adversarial example x^{adv} for some model such that $\|x^{\text{adv}} - x\|_\infty \leq \epsilon$, and outputting $(x^{\text{adv}}, y_{\text{true}})$. In Ensemble Adversarial Training, the source distributions are $\mathcal{D}$ (the clean data) and $\mathcal{A}_1, \ldots, \mathcal{A}_k$ (the attacks overs the currently trained model and the static pre-trained models). The target distribution takes the form of an unseen black-box adversary $\mathcal{A}^*$. Standard generalization bounds for Domain Adaptation (Mansour et al., 2009; Zhang et al., 2012) yield the following result.

Theorem 1 (informal)**.** *Let $h^* \in \mathcal{H}$ be a model learned with Ensemble Adversarial Training and static black-box adversaries $\mathcal{A}_1, \ldots, \mathcal{A}_k$. Then, if h^* is robust against the black-box adversaries $\mathcal{A}_1, \ldots \mathcal{A}_k$ used at training time, then h^* has bounded error on attacks from a future black-box adversary $\mathcal{A}^*$, if $\mathcal{A}^*$ is not "much stronger", on average, than the static adversaries $\mathcal{A}_1, \ldots, \mathcal{A}_k$.*

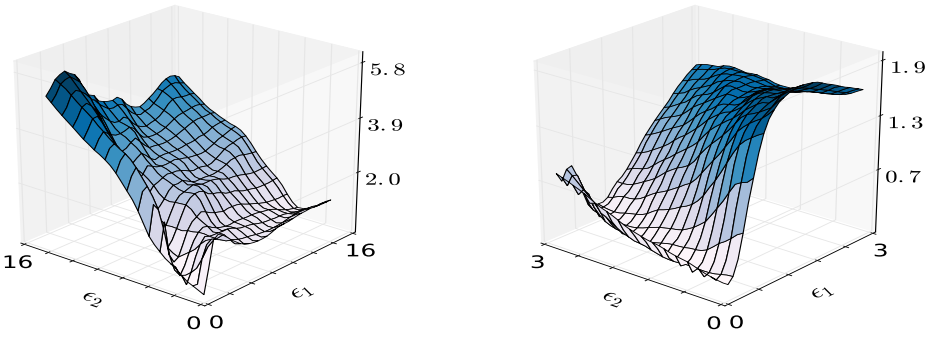
We give a formal statement of this result and of the assumptions on $\mathcal{A}^*$ in Appendix B. Of course, ideally we would like guarantees against *arbitrary* future adversaries. For very low-dimensional tasks (e.g., MNIST), stronger guarantees are within reach for specific classes of adversaries (e.g., ℓ_∞ bounded perturbations (Madry et al., 2017; Sinha et al., 2018; Raghunathan et al., 2018; Kolter & Wong, 2017)), yet they also fail to extend to other adversaries not considered at training time (see Appendix C.1 for a discussion). For ImageNet-scale tasks, stronger formal guarantees appear out of reach, and we thus resort to an experimental assessment of the robustness of Ensemble Adversarially Trained models to various non-interactive black-box adversaries in Section 4.2.[2]

4 EXPERIMENTS

We show the existence of a degenerate minimum, as described in Section 3.3, for the adversarially trained Inception v3 model of Kurakin et al. (2017b). Their model (denoted v3_{adv}) was trained on a Step-LL attack with $\epsilon \leq 16/256$. We also adversarially train an Inception ResNet v2 model (Szegedy et al., 2016a) using the same setup. We denote this model by IRv2_{adv}. We refer the reader to (Kurakin et al., 2017b) for details on the adversarial training procedure.

We first measure the *approximation-ratio* of the Step-LL attack for the inner maximization in (1). As we do not know the true maximum, we lower-bound it using an iterative attack. For $1{,}000$ random test points, we find that for a standard Inception v3 model, step-LL gets within 19% of the optimum loss on average. This attack is thus a good candidate for adversarial training. Yet, for the v3_{adv} model, the approximation ratio drops to 7%, confirming that the learned model is less amenable to linearization. We obtain similar results for Inception ResNet v2 models. The ratio is 17% for a standard model, and 8% for IRv2_{adv}. Similarly, we look at the *cosine similarity* between the perturbations given by a single-step and multi-step attack. The more linear the model, the more similar we expect both perturbations to be. The average similarity drops from 0.13 for Inception v3

[2]We note that subsequent work did succeed in scaling both empirical (Xie et al., 2019a) and certified (Cohen et al., 2019) white-box defenses to ImageNet-scale tasks.



(a) Loss of model $v3_{adv}$.

(b) Zoom in for small ϵ_1, ϵ_2.

Figure 1: **Gradient masking in single-step adversarial training.** We plot the loss of model $v3_{adv}$ on points $x^* = x + \epsilon_1 \cdot g + \epsilon_2 \cdot g^\perp$, where g is the signed gradient and $g^\perp$ is an orthogonal adversarial direction. Plot (b) is a zoom of (a) near x. The gradient poorly approximates the global loss.

to 0.02 for $v3_{adv}$. This effect is not due to the decision surface of $v3_{adv}$ being "too flat" near the data points: the average gradient norm is larger for $v3_{adv}$ (0.17) than for the standard v3 model (0.10).

We visualize this "gradient-masking" effect (Papernot et al., 2016b) by plotting the loss of $v3_{adv}$ on examples $x^* = x + \epsilon_1 \cdot g + \epsilon_2 \cdot g^\perp$, where g is the signed gradient of model $v3_{adv}$ and $g^\perp$ is a signed vector orthogonal to g. Looking forward to Section 4.1, we actually chose $g^\perp$ to be the signed gradient of another Inception model, from which adversarial examples transfer to $v3_{adv}$. Figure 1 shows that the loss is highly curved in the vicinity of the data point x, and that the gradient poorly reflects the global loss landscape. Similar plots for additional data points are in Figure 4.

We show similar results for adversarially trained MNIST models in Appendix C.2. On this task, *input dropout* (Srivastava et al., 2014) mitigates adversarial training's overfitting problem, in some cases. Presumably, the random input mask *diversifies* the perturbations seen during training (dropout at intermediate layers does not mitigate the overfitting effect). Mishkin et al. (2017) find that input dropout significantly degrades accuracy on ImageNet, so we did not include it in our experiments.

4.1 Attacks Against Adversarially Trained Networks

Kurakin et al. (2017b) found their adversarially trained model to be robust to various single-step attacks. They conclude that this robustness should translate to attacks *transferred* from other models. As we have shown, the robustness to single-step attacks is actually misleading, as the model has learned to *degrade* the information contained in the model's gradient. As a consequence, we find that the $v3_{adv}$ model is substantially more vulnerable to single-step attacks than Kurakin et al. (2017b) predicted, both in a white-box and black-box setting. The same holds for the $IRv2_{adv}$ model.

In addition to the $v3_{adv}$ and $IRv2_{adv}$ models, we consider standard Inception v3, Inception v4 and Inception ResNet v2 models. These models are available in the `TensorFlow-Slim` library (Abadi et al., 2015). We describe similar results for a variety of models trained on MNIST in Appendix C.2.

Black-box attacks. Table 1 shows error rates for single-step attacks transferred between models. We compute perturbations on one model (the source) and transfer them to all others (the targets). When the source and target are the same, the attack is white-box. Adversarial training greatly increases robustness to white-box single-step attacks, but incurs a higher error rate in a black-box setting. Thus, the robustness gain observed when evaluating defended models in isolation is misleading. Given the ubiquity of this pitfall among proposed defenses against adversarial examples (Carlini & Wagner, 2017a; Brendel & Bethge, 2017; Papernot et al., 2016b), we advise researchers *to always consider both white-box and black-box adversaries when evaluating defensive strategies*. Notably, a similar discrepancy between white-box and black-box attacks was recently observed in Buckman et al. (2018).

Attacks crafted on adversarial models are found to be weaker even against undefended models (i.e., when using $v3_{adv}$ or $IRv2_{adv}$ as source, the attack transfers with lower probability). This confirms